

ST CROIX ROHLSSEN, Virgin Islands, U.S.

WMO: 785510

Lat: 17.700N

Lon: 64.813W

Elev: 61

StdP: 14.66

Time Zone: -4.00 (W04)

Period: 00-19

WBAN: 11624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.9	69.5	60.3	78.2	75.4	61.8	82.7	75.8	23.0	81.0	21.6	81.3	5.4	0	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	89.8	78.5	89.1	78.3	88.4	78.1	80.6	87.2	80.0	86.7	79.5	86.3	16.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	148.4	84.2	77.9	145.3	84.0	77.3	142.5	83.9	44.1	87.3	43.4	86.9	42.9	86.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.8	20.3	18.8	DB	63.5	93.3	5.5	6.0	59.5	97.6	56.3	101.1	53.3	104.5	49.3	108.8
			WB	62.0	81.9	1.7	1.5	60.7	82.9	59.7	83.8	58.7	84.7	57.5	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	77.4	77.9	78.3	79.6	80.8	82.8	83.4	83.5	82.8	82.1	80.2	79.0
	DBStd	2.85	2.02	1.91	1.94	1.81	1.97	1.75	1.50	1.49	1.95	2.10	2.05	2.03
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11196	851	782	876	887	956	985	1035	1038	984	997	907	898
	CDD65	5721	386	362	411	437	491	535	570	573	534	532	457	433
	CDH74	56288	2601	2590	3149	3893	4918	6114	6636	6771	6149	5790	4216	3461
	CDH80	16870	438	504	647	935	1311	2086	2379	2532	2256	1974	1111	697
Wind	WSAvg	10.5	10.4	11.1	10.5	10.4	10.9	12.1	12.6	11.2	8.8	8.6	8.6	10.4
Precipitation	PrecAvg	42.2	2.2	1.9	1.9	2.6	4.0	2.7	3.1	4.1	5.3	5.2	5.4	3.9
	PrecMax	78.9	7.4	3.6	4.1	8.5	17.9	7.8	6.9	12.0	21.4	16.2	16.3	12.9
	PrecMin	27.2	0.6	0.2	0.4	0.0	0.2	0.1	0.5	1.4	1.1	1.1	1.1	0.8
	PrecStd	10.9	1.3	1.0	1.0	2.1	4.0	1.9	1.7	2.6	3.9	3.5	3.9	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	86.4	86.4	87.3	88.1	89.7	91.3	91.4	91.9	89.9	88.5	87.2
		MCWB	73.9	73.6	74.9	76.1	77.4	78.1	78.4	79.2	79.8	78.6	77.7	76.1
	2%	DB	84.5	85.2	85.4	86.3	87.3	88.7	89.5	89.6	89.4	88.9	87.4	86.1
		MCWB	74.0	73.8	74.6	75.5	77.1	77.9	78.1	78.6	78.8	78.3	77.4	75.8
	5%	DB	83.8	84.3	84.6	85.5	86.4	88.0	88.6	88.8	88.3	88.0	86.4	85.0
		MCWB	73.9	73.6	74.1	75.2	76.8	77.7	77.9	78.4	78.6	78.1	77.1	75.4
	10%	DB	83.0	83.4	83.8	84.6	85.5	87.3	87.7	87.9	87.6	87.1	85.4	84.0
		MCWB	73.6	73.2	73.7	74.9	76.6	77.6	77.7	78.2	78.5	78.0	76.8	75.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	77.4	78.0	78.9	79.8	80.7	80.8	81.0	81.3	80.8	79.9	78.1
		MCDB	82.1	82.0	82.9	84.8	85.7	86.8	87.2	87.8	90.9	88.3	85.7	83.8
	2%	WB	75.9	75.6	76.4	77.5	78.6	79.7	79.9	80.1	80.2	79.9	78.8	77.1
		MCDB	81.5	81.9	82.5	83.8	85.0	86.3	86.7	87.1	87.2	86.7	85.1	83.4
	5%	WB	75.1	74.7	75.5	76.7	77.9	78.9	79.2	79.7	79.7	79.2	78.1	76.3
		MCDB	81.0	81.5	81.8	83.1	84.2	85.9	86.3	86.6	86.8	86.1	84.2	82.6
	10%	WB	74.3	73.9	74.6	75.8	77.2	78.2	78.5	79.1	79.1	78.7	77.4	75.5
		MCDB	80.3	80.8	81.2	82.3	83.5	85.1	85.6	86.0	86.2	85.6	83.3	81.7
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.2	11.2	10.6	9.7	9.5	9.9	10.1	10.7	10.6	10.6	10.4
		MCDBR	11.4	11.9	11.4	11.1	10.6	9.9	10.5	10.7	11.3	11.2	11.4	11.1
		MCWBR	5.1	5.0	4.9	4.5	4.3	4.0	4.2	4.3	4.7	4.7	4.9	4.9
	5% WB	MCDBR	11.0	11.1	10.5	10.1	9.2	9.5	9.9	10.1	10.9	10.5	10.3	10.1
MCWBR		5.4	5.2	5.1	4.7	4.4	4.4	4.5	4.6	5.0	4.9	5.1	5.0	
Clear-Sky Solar Irradiance	taub	0.369	0.375	0.380	0.405	0.448	0.513	0.517	0.477	0.437	0.392	0.378	0.374	
	taud	2.583	2.561	2.544	2.497	2.387	2.165	2.149	2.300	2.442	2.587	2.620	2.606	
	Ebn at Noon	291	295	295	286	270	251	250	263	274	286	287	286	
	Edh at Noon	30	32	34	35	39	48	49	43	36	31	29	29	
All-Sky Solar Radiation	RadAvg	1568	1824	2010	2104	2030	2042	2077	2046	1912	1693	1500	1453	
	RadStd	37	34	69	108	124	115	92	91	117	114	95	65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.41	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	+142	+144

Nomenclature: See separate page